

7.1

Vďaka vlastnostiam mocnenia matic v Jordanovej forme môžeme maticu po blokoch:

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 \\ 0 \end{pmatrix} \rightarrow$$

$$\rightarrow O_{5 \times 5}$$

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow O_{3 \times 3} ; \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \rightarrow O_{2 \times 2}$$

Pre $k \geq 5$ je teda matica zobrazenia f^k nulová, teda pre $i \geq 5$ je $\text{Ker } f^i = \mathbb{R}^{10}$

a pre $j \geq 5$ je $\text{Im } f^j = \vec{0}$

Teraz poďme postupne:

Pre $k=4$ je matica $[f^k]_B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, teda $\text{Ker } f^k = \mathbb{R}^{10} \setminus \text{span}\{b_5\}$ a $\text{Im } f^k = \text{span}\{b_1\}$

Pre $k=3$ je matica $[f^k]_B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, teda $\text{Ker } f^k = \mathbb{R}^{10} \setminus \text{span}\{b_5, b_9\}$ a $\text{Im } f^k = \text{span}\{b_1, b_2\}$

Pre $k=2$ $[f^k]_B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, teda $\text{Ker } f^k = \text{span}\{b_1, b_2, b_6, b_7, b_9, b_{10}\}$ a $\text{Im } f^k = \text{span}\{b_1, b_2, b_3, b_6\}$

Pre $k=1$ $[f^k]_B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$, teda $\text{Ker } f^k = \text{span}\{b_1, b_6, b_9\}$ a $\text{Im } f^k = \text{span}\{b_1, b_2, b_3, b_4, b_6, b_7, b_9\}$

Môžeme teda postupne rozobrať všetky možnosti:

• $j \geq 5$: $\text{Im } f^j = \vec{0}$, teda $\text{Ker } f^i \cap \text{Im } f^j = \vec{0} \rightarrow \text{dimenzia } 0$

• $j=4$: $\text{Im } f^j = \text{span}\{b_1\}$ a pre všetky i patrí $\text{span}\{b_1\}$ do $\text{Ker } f^i$ teda $\text{Ker } f^i \cap \text{Im } f^j = \text{span}\{b_1\} \rightarrow \text{dimenzia } 1$ s bázou $B = (b_1)$.

• $j=3$: $\text{Im } f^j = \text{span}\{b_1, b_2\}$ a pre vsetky iz 2 plati $\text{Ker } f^i \cap \text{Im } f^j = \text{span}\{b_1, b_2\}$ teda dimenzie 2 s bazon $B = (b_1, b_2)$.

$\text{Pre } i=1$ je $\text{Ker } f^i \cap \text{Im } f^j = \text{span}\{b_1\}$ dimenzie 1 s bazon $B = (b_1)$.

• $j=2$: $\text{Im } f^j = \text{span}\{b_1, b_2, b_3, b_6\}$, teda pre $i \geq 3$ plati $\text{Ker } f^i \cap \text{Im } f^j = \text{span}\{b_1, b_2, b_3, b_6\}$ dimenzie 4 s bazon $B = (b_1, b_2, b_3, b_6)$

$\text{Pre } i=2$ je to $\text{span}\{b_1, b_2, b_6\}$ dimenzie 3 s bazon $B = (b_1, b_2, b_6)$

$\text{Pre } i=1$ je to $\text{span}\{b_1, b_6\}$ dimenzie 2 s bazon $B = (b_1, b_6)$

• $j=1$:
 $\text{Im } f^j = \text{span}\{b_1, b_2, b_3, b_4, b_6, b_7, b_9\}$ pre $i \geq 4$ plati $\text{Ker } f^i \cap \text{Im } f^j = \text{Im } f^j$ dimenzie 7 s bazon $B = (b_1, b_2, b_3, b_4, b_6, b_7, b_9)$

$\text{Pre } i=3$ je to $\text{span}\{b_1, b_2, b_3, b_6, b_7, b_9\}$ dimenzie 6 s bazon $B = (b_1, b_2, b_3, b_6, b_7, b_9)$

$\text{Pre } i=2$ je to $\text{span}\{b_1, b_2, b_6, b_7, b_9\}$ dimenzie 5 s bazon $B = (b_1, b_2, b_6, b_7, b_9)$

$\text{Pre } i=1$ je to $\text{span}\{b_1, b_6, b_9\}$ dimenzie 3 s bazon $B = (b_1, b_6, b_9)$